

| Question | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
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| 7        | <p><b>Indicative content</b></p> <p><b>Feedback loop</b><br/> A1 Detector {sensitive / reacts} to change in {set / normal} {point / level}<br/> A2 Coordinator receives input from detector<br/> A3 and responds by sending an impulse to effector<br/> A4 Effector bring about a change that corrects the deviation / owtte</p> <p><b>Osmoregulation</b><br/> B1 {Hypothalamus / osmoreceptors} detects change in {water potential / body fluid concentration / concentration of blood}<br/> B2 Hypothalamus sends a signal to the (posterior) pituitary gland<br/> B3 Posterior (pituitary gland) releases (more) ADH if {water potential decreases / body fluid concentration increases} / ORA<br/> B4 Walls of DCT and collecting ducts act as effectors<br/> B5 ADH makes walls of {DCT / collecting duct} more permeable to water Accept more aquaporins incorporated into membranes of {DCT / CD} / ORA<br/> B6 <u>More</u> water {diffuses out of collecting duct / reabsorbed} into {medulla / blood} / ORA</p> <p><b>Excessive alcohol use</b><br/> C1 it is more difficult for fluid to pass through / less ultrafiltration occurs / ultrafiltration is more difficult<br/> C2 Less fluid will leave the blood / more water remains in the blood / less filtrate is formed<br/> C3 Blood is {more dilute / less concentrated} / blood has a higher water potential<br/> C4 Fluid will accumulate {in tissues / reference to oedema} / stops water being drawn back into the blood</p> |                 |     |     |       |       |      |